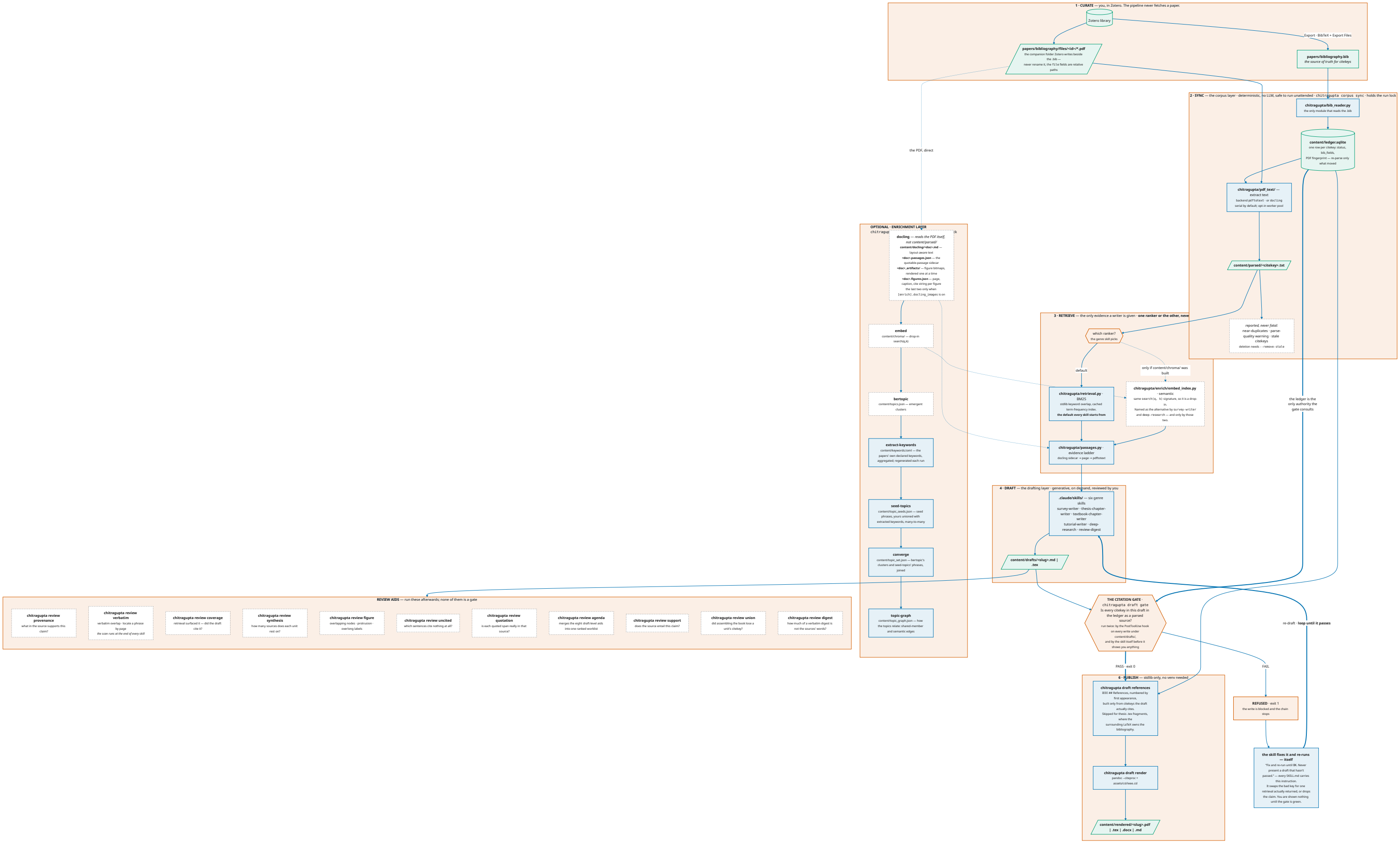
The full workflow — every module, every artifact


1 · CURATE — you, in Zotero. The pipeline never fetches a paper.
Zotero library
Export · BibTeX + Export Files
papers/bibliography/files/<id>/*.pdf the companion folder Zotero writes beside the .bib — never rename it, the `file` fields are relative paths
papers/bibliography.bib *the source of truth for citekeys*
the PDF, direct
2 · SYNC — the corpus layer · deterministic, no LLM, safe to run unattended · `chitragupta corpus sync` · holds the run lock
chitragupta/bib_reader.py the only module that reads the .bib
content/ledger.sqlite one row per citekey: status, bib_fields, PDF fingerprint — re-parse only what moved
chitragupta/pdf_text/ — extract text backend `pdftotext` · or `docling` serial by default; opt-in worker pool
content/parsed/<citekey>.txt
reported, never fatal: near-duplicates · parse-quality warning · stale citekeys deletion needs `--remove-stale`
the ledger is the only authority the gate consults
OPTIONAL · ENRICHMENT LAYER
`chitragup` … ck
docling — *reads the PDF itself, not content/parsed/* **content/docling/<doc>.md** — layout-aware text **<doc>.passages.json** — the quotable-passage sidecar **<doc>_artifacts/** — figure bitmaps, rendered one at a time **<doc>.figures.json** — page, caption, cite string per figure the last two only when `[enrich].docling_images` is on
embed content/chroma/ — drop-in search(q,k)
bertopic content/topics.json — emergent clusters
extract-keywords content/keywords.toml — the papers' own declared keywords, aggregated; regenerated each run
seed-topics content/topic_seeds.json — seed phrases, yours unioned with extracted keywords, many-to-many
converge content/topic_set.json — bertopic's clusters and seed-topics' phrases, joined
topic-graph content/topic_graph.json — how the topics relate: shared-member and semantic edges
3 · RETRIEVE — the only evidence a writer is given · **one ranker or the other, neve**
which ranker? the genre skill picks
default
only if content/chroma/ was built
chitragupta/retrieval.py · BM25 stdlib keyword overlap, cached term-frequency index. **the default every skill starts from**
chitragupta/enrich/embed_index.py · semantic same `search(q, k)` signature, so it is a drop-in. Named as the alternative by `survey-writer` and `deep-research` — and only by those two.
chitragupta/passages.py · evidence ladder docling sidecar → page → pdftotext
4 · DRAFT — the drafting layer · generative, on demand, reviewed by you
.claude/skills/ — six genre skills survey-writer · thesis-chapter-writer · textbook-chapter-writer tutorial-writer · deep-research · review-digest
content/drafts/<slug>.md | .tex
THE CITATION GATE · `chitragupta draft gate` Is every citekey in this draft in the ledger as a parsed source? run twice: by the PostToolUse hook on every write under content/drafts/, and by the skill itself before it shows you anything
PASS · exit 0
FAIL
re-draft · **loop until it passes**
6 · PUBLISH — stdlib only, no venv needed
chitragupta draft references IEEE ## References, numbered by first appearance, built only from citekeys the draft actually cites. Skipped for thesis .tex fragments, where the surrounding LaTeX owns the bibliography.
chitragupta draft render pandoc --citeproc + assets/csl/ieee.csl
content/rendered/<slug>.pdf | .tex | .docx | .md
REFUSED · exit 1 the write is blocked and the chain stops
the skill fixes it and re-runs — itself "Fix and re-run until 0K. Never present a draft that hasn't passed." — every SKILL.md carries this instruction. It swaps the bad key for one retrieval actually returned, or drops the claim. You are shown nothing until the gate is green.
REVIEW AIDS — run these afterwards; none of them is a gate
chitragupta review provenance what in the source supports this claim?
chitragupta review verbatim verbatim overlap · locate a phrase by page *the scan runs at the end of every skill*
chitragupta review coverage retrieval surfaced it — did the draft cite it?
chitragupta review synthesis how many sources does each unit rest on?
chitragupta review figure overlapping nodes · protrusion · overlong labels
chitragupta review uncited which sentences cite nothing at all?
chitragupta review quotation is each quoted span really in that source?
chitragupta review agenda merges the eight draft-level aids into one ranked worklist
chitragupta review support does the source entail this claim?
chitragupta review union did assembling the book lose a unit's citekey?
chitragupta review digest how much of a verbatim digest is not the sources' words?